

Lecture 10: Second derivative test: boundaries and infinity

How do we decide between $\begin{cases} \text{local min/max} \\ \text{local max/min} \\ \text{saddle point} \end{cases}$

How do we find global min/max?

These occur either at a critical point or on boundary/at infinity

Second derivative test: First consider $w = ax^2 + by^2 + cy^2$

Example: $w = x^2 + 2xy + 3y^2 = (x+y)^2 + 2y^2$

In general, if $a \neq 0$ $w = a\left(x + \frac{b}{2a}y\right) + cy^2 \Leftrightarrow w = a\left(x + \frac{b}{2a}y\right) + \left(c - \frac{b^2}{4a}\right)y^2 \Leftrightarrow w = \frac{1}{4a} \left[4a^2 \left(x + \frac{b}{2a}y\right)^2 + (4ac - b^2)y^2 \right]$

3 cases: 1) $4ac - b^2 < 0 \Rightarrow$ Saddle point one term > 0 , the other < 0

2) $4ac - b^2 = 0$ degenerate critical point

3) $4ac - b^2 > 0$, $w = \frac{1}{4a} \left[4 \left(\dots \right)^2 + \left(\dots \right)^2 \right] \geq 0$

$\rightarrow$ if $a > 0$: minimum

$\rightarrow$ if $a < 0$: maximum

$b^2 - 4ac$: quadratic formula

$$w = ax^2 + by^2 + cy^2 = \frac{1}{4a} \left[a \left(\frac{x}{a} \right)^2 + b \left(\frac{y}{a} \right)^2 + c \right]$$

if $b^2 - 4ac > 0$ then this takes $\oplus$ and $\ominus$ values

w takes positive and negative values $\Rightarrow$ saddle point

If $b^2 - 4ac < 0$ $a \left(\frac{x}{a} \right)^2 + b \left(\frac{y}{a} \right)^2 + c$ always ≥ 0 or ≤ 0

In general: look at second derivatives! $\frac{\partial^2 f}{\partial x^2} = f_{xx}$ $\frac{\partial^2 f}{\partial x \partial y} = f_{xy}$ $\frac{\partial^2 f}{\partial y \partial x} = f_{yx}$ $\frac{\partial^2 f}{\partial y^2} = f_{yy}$

Second derivative test: At a critical point (x_0, y_0) of f

Let $A = f_{xx}(x_0, y_0)$, $B = f_{xy}(x_0, y_0)$, $C = f_{yy}(x_0, y_0)$

If $AC - B^2 > 0$, and $A > 0$, local minimum; $AC - B^2 > 0$ and $A < 0$, local maximum; $AC - B^2 < 0$: Saddle; $AC - B^2 = 0$: can't conclude

*Verify: in special case $w = ax^2 + by^2 + cy^2$

$$w_x = 2ax + by$$

$$w_y = bx + 2cy$$

$$w_x = w_y = 0$$

$$w_{xx} = 2a$$

$$w_{yy} = 2c$$

$$A = 2a; B = b; C = 2c \quad AC - B^2 = 4ac - b^2$$

$$\text{Quadratic approximation: } \Delta f \approx \underbrace{f_x(x_0, y_0)}_{=0 \text{ at } P} (x-x_0) + \underbrace{f_y(x_0, y_0)}_{=0 \text{ at } P} (y-y_0) + \frac{1}{2} f_{xx}(x_0, y_0) (x-x_0)^2 + f_{xy}(x_0, y_0) (x-x_0)(y-y_0) + \frac{1}{2} f_{yy}(x_0, y_0) (y-y_0)^2$$

$\rightarrow$ so the general case reduces to quadratic case. In degenerate case, depends actually on higher order derivatives

Example: $f(x, y) = x^2y + \frac{1}{2}xy^2$ $x, y > 0$ Critical points? $f_x = x^2 - 1 = 0$, $f_y = x - \frac{1}{2} = 0$

$$\begin{cases} x^2 = 1 & x = y = 1 \\ x^2 = 1 & y = 1 \end{cases}$$

$$f_{xx} = \frac{2}{x^2} \quad A = 2 \quad f_{xy} = \frac{1}{x^2} \quad B = 1 \quad f_{yy} = \frac{2}{x^2} \quad C = 2$$

$AC - B^2 = 4 - 1 = 3 > 0$ and $A > 0$ so local minimum

Max/min: $f \rightarrow \infty$ when $x \rightarrow \infty$ or $y \rightarrow \infty$ or $x, y \rightarrow 0$